Continental AG
BLE Tire Sensor System

Technical Reference Manual v2.1

Internal Use Only — Tire Sensor Analytics Division
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1. System Overview

The Continental BLE Tire Sensor System (TSS) is a wireless monitoring platform designed for real-time acquisition of tire performance data. Each sensor node is embedded inside the tire inner-liner and communicates via Bluetooth Low Energy (BLE 5.0) to a vehicle-mounted gateway. The system supports up to 24 simultaneous sensor connections per gateway, enabling fleet-wide monitoring.

Primary use cases include predictive maintenance, fuel efficiency optimization, and safety-critical alerts for tire blowout prevention. The sensor firmware is updated over-the-air (OTA) through the gateway management interface.

2. Sensor Types and Variants

Continental deploys three sensor variants depending on the application domain:

Variant	Model	Measurements	Sampling Rate	Battery Life
Standard	TSS-100	Pressure, Temperature	1 Hz	5 years
Advanced	TSS-200	Pressure, Temp, Tread Depth	5 Hz	3 years
Racing	TSS-300R	Pressure, Temp, Tread, Acceleration (3-axis)	50 Hz	6 months

The TSS-200 is the most widely deployed variant, used across Continental's commercial tire division. It features an integrated MEMS pressure transducer (range 0-8 bar, accuracy +/- 0.05 bar) and a thermistor-based temperature sensor (range -40 to 150 deg C, accuracy +/- 1 deg C). Tread depth is measured via an ultrasonic time-of-flight sensor with 0.1mm resolution.

3. BLE Data Packet Format

Each sensor transmits data as a 20-byte BLE advertisement packet. The packet structure is as follows:

Byte Offset	Length	Field	Description	Encoding
0x00	2	Header	Packet identifier	0xCO 0x4E (magic bytes)
0x02	4	Sensor ID	Unique sensor identifier	uint32, big-endian
0x06	2	Pressure	Tire pressure in mbar	uint16, big-endian
0x08	2	Temperature	Temperature in 0.1 deg C	int16, big-endian
0x0A	2	Tread Depth	Tread in 0.01mm units	uint16, big-endian
0x0C	4	Timestamp	Unix epoch seconds	uint32, big-endian
0x10	1	Battery	Battery level percentage	uint8
0x11	1	Status	Status flags	Bitfield (see 3.1)
0x12	2	CRC-16	Packet checksum	CRC-CCITT

3.1 Status Flags (Byte 0x11)

Bit 0: Sensor OK (1=healthy). Bit 1: Low battery warning. Bit 2: Pressure alert (under-inflation detected). Bit 3: Temperature alert (overheating). Bit 4: Tread alert (minimum depth reached). Bits 5-7: Reserved for future use.

Example raw packet (hex): CO 4E 00 0A 3F 21 09 C4 01 A4 1F 40 67 8B 2C 10 5E 04 A3 B7. Decoded: Sensor ID = 0x000A3F21 (671521), Pressure = 2500 mbar (2.5 bar), Temperature = 42.0 deg C, Tread = 80.00mm, Battery = 94%.

4. Measurement Ranges and Thresholds

The following operational thresholds are configured by default. Fleet operators can customize thresholds via the gateway management API.

Parameter	Normal Range	Warning	Critical Alert
Tire Pressure	2.0 - 3.5 bar	< 1.8 bar or > 3.8 bar	< 1.5 bar or > 4.2 bar
Temperature	-20 to 80 deg C	> 90 deg C	> 110 deg C
Tread Depth	8.0 - 3.0 mm	< 3.0 mm	< 1.6 mm (legal limit)
Battery Level	100% - 20%	< 20%	< 5%

When a critical alert is triggered, the gateway immediately pushes a notification to the fleet management dashboard and logs the event to the AWS S3 data lake. The alert includes sensor ID, measurement value, threshold exceeded, GPS coordinates, and timestamp.

5. Data Pipeline Architecture

The sensor data flows through the following pipeline:

1. **Acquisition**: BLE sensor broadcasts raw packet every 1-50 Hz depending on variant.
2. **Gateway Processing**: Vehicle gateway receives packets, decodes hex payload, validates CRC-16 checksum, and buffers data locally.
3. **Cloud Ingestion**: Buffered data is uploaded to AWS S3 via MQTT over cellular connection. Each upload contains a JSON batch of decoded sensor readings.
4. **Storage**: Raw JSON is stored in S3 partitioned by date and fleet ID. A Glue ETL job transforms data into Parquet format for analytics.
5. **Analytics**: Streamlit dashboards and Jupyter notebooks query the Parquet data via Athena for fleet health monitoring and predictive maintenance.

6. Troubleshooting Guide

Symptom	Possible Cause	Resolution
No BLE signal	Sensor battery depleted	Replace sensor module
No BLE signal	Gateway firmware outdated	OTA update to v4.2+
Incorrect pressure reading	Sensor calibration drift	Run recalibration via gateway CLI
Intermittent data loss	BLE interference (2.4 GHz)	Relocate gateway antenna
High temperature alert (false)	Sensor near brake assembly	Reposition sensor on inner-liner
CRC checksum failure	Packet corruption	Check BLE signal strength (RSSI > -70 dBm)

7. Glossary

BLE: Bluetooth Low Energy — wireless protocol for short-range sensor communication.
MEMS: Micro-Electro-Mechanical Systems — miniaturized pressure transducer technology.
RSSI: Received Signal Strength Indicator — BLE signal quality metric in dBm.
CRC-16: Cyclic Redundancy Check — 16-bit checksum for packet integrity validation.
OTA: Over-The-Air — wireless firmware update mechanism.
MQTT: Message Queuing Telemetry Transport — lightweight IoT messaging protocol.
Gateway: Vehicle-mounted BLE receiver that aggregates sensor data and uploads to cloud.
Inner-liner: Interior rubber layer of a tire where sensors are bonded.
Parquet: Columnar storage format optimized for analytical queries.
Athena: AWS serverless query engine for S3-based data lakes.